

998285

Let G denote the gcd of the integers

$$\frac{P}{a_i} + 1 \quad (1 \leq i \leq k).$$

0.0.1 Upper bound

For every i ,

$$\frac{P}{a_i} \equiv -1 \pmod{G}.$$

Multiplying by a_i gives

$$P \equiv -a_i \pmod{G},$$

so all the a_i are congruent modulo G .

Suppose $G > 998285$. In particular $G > 1$. No a_i is divisible by a prime divisor of G : for $j \neq i$, the number a_i divides P/a_j , while

$$\gcd\left(\frac{P}{a_j}, G\right) = 1.$$

Thus their common residue is nonzero modulo G . Write it as c , where

$$1 \leq c < G.$$

After ordering the a_i , they have the form

$$a_i = c + n_i G$$

for distinct nonnegative integers n_i . Hence $n_i \geq i - 1$, and therefore

$$\sum_{i=1}^k a_i \geq kc + \frac{k(k-1)}{2}G.$$

If $k \geq 4$, then

$$\sum_{i=1}^k a_i \geq 6G > 6 \cdot 998285 > 3 \cdot 10^6,$$

which is impossible. Thus any value exceeding 998285 would have to occur when $k = 3$.

For $k = 3$, the preceding bound gives

$$3c + 3G < 3 \cdot 10^6,$$

so

$$c + G \leq 999999.$$

Since the three a_i are all congruent to $c \pmod{G}$,

$$\frac{P}{a_i} \equiv c^2 \pmod{G}.$$

But $P/a_i \equiv -1 \pmod{G}$, so

$$c^2 \equiv -1 \pmod{G}.$$

Consequently,

$$c^2 + 1 = mG$$

for some positive integer m .

Our assumption $G > 998285$, together with 1, gives

$$G \geq 998286, \quad c \leq 1713.$$

Thus

$$c^2 + 1 \leq 1713^2 + 1 = 2934370 < 3 \cdot 998286 \leq 3G.$$

Equation 2 therefore forces $m = 1$ or $m = 2$.

If $m = 1$, then $G = c^2 + 1$. From 1,

$$c^2 + c + 1 \leq 999999.$$

Since $1000^2 + 1000 + 1 > 999999$, we have $c \leq 999$, and hence

$$G \leq 999^2 + 1 = 998002,$$

contradicting $G > 998285$.

If $m = 2$, then

$$G = \frac{c^2 + 1}{2}.$$

Using 1,

$$\frac{c^2 + 1}{2} + c \leq 999999,$$

or equivalently

$$(c + 1)^2 \leq 1999998.$$

Because

$$1415^2 = 2002225 > 1999998,$$

we obtain $c \leq 1413$. Therefore

$$G \leq \frac{1413^2 + 1}{2} = \frac{1996570}{2} = 998285,$$

again contradicting $G > 998285$.

Hence in every admissible configuration,

$$G \leq 998285.$$

0.0.2 Construction attaining the bound

Set

$$c = 1413, \quad G = 998285.$$

Then

$$c^2 + 1 = 1413^2 + 1 = 1996570 = 2G.$$

Take

$$a_1 = c = 1413, \quad a_2 = c + G = 999698, \quad a_3 = c + 2G = 1997983.$$

These are distinct positive integers, and

$$a_1 + a_2 + a_3 = 2999094 < 3000000.$$

All three numbers are congruent to $c \pmod{G}$, and $c^2 \equiv -1 \pmod{G}$. Therefore each pairwise product plus 1, and hence each $P/a_i + 1$, is divisible by G .

It remains to check that their common gcd is not larger. We have

$$a_1a_2 + 1 = c(c + G) + 1 = G(c + 2) = 1415G,$$

and

$$a_1a_3 + 1 = c(c + 2G) + 1 = G(2c + 2) = 2828G.$$

But

$$\gcd(1415, 2828) = 1,$$

since $2828 - 2 \cdot 1415 = -2$ and 1415 is odd. Thus

$$\gcd(a_1a_2 + 1, a_1a_3 + 1) = G.$$

As all three required numbers are divisible by G , their gcd is exactly G .

Therefore the maximum possible value is

$$\boxed{998285}.$$